# Coordinate geometry: lines, projections and circles

## Lecture and worked examples

Maths Explorer

## Chapter path

**Objectives.** Use a normal vector; calculate a projection; determine or recognise a circle; translate a configuration into equations.

**Prerequisites.** Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

**Yearly reference point.** 5, 7 and 12 January: 6 hours

## Normal vector and Cartesian equation

A nonzero normal vector to a line is orthogonal to every direction vector of that line. If $\vec{n} = (a, b) \neq (0, 0)$ is normal and $A(x_A, y_A)$ belongs to d, then $d$ has equation $a(x - x_A) + b(y - y_A) = 0$, that is $ax + by + c = 0$. A possible direction vector is $(-b, a)$.

## Normal vector and Cartesian equation

Through $A(1, 2)$ with normal $(2, -3)$: $2(x - 1) - 3(y - 2) = 0$, so $2x - 3y + 4 = 0$. The vector $(3, 2)$ is a direction vector of the line: scalar product $6 - 6 = 0$.

## Normal vector and Cartesian equation

$(a, b)$ is normal, not a direction vector in general. $(a, b)$ must not be zero.

## Normal vector and Cartesian equation

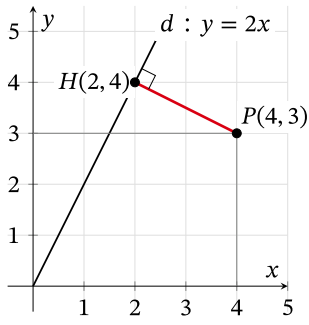
Equation of the line through $(2, -1)$ with normal $(3, 4)$.

## Orthogonal projection and distance

The projection H of P onto d is characterised by $H \in d$ and $\overrightarrow{PH}$ collinear with a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a, b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. The distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a derived extension rather than a standalone routine.

# Orthogonal projection and distance

For $P(3,4)$ and $d: x+y-2=0$,
$\lambda = \dfrac{3+4-2}{2} = \dfrac{5}{2}$, so $H(\dfrac{1}{2}, \dfrac{3}{2})$ and
$PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.



## Orthogonal projection and distance

The distance is nonnegative: keep the absolute value. Two checks, membership and perpendicularity, verify the projection.

## Orthogonal projection and distance

Project $P(4, 1)$ onto $d : x = 2$.

## Equation of a circle

The circle with centre $I(\alpha, \beta)$ and radius $r > 0$ has equation $(x-\alpha)^2 + (y-\beta)^2 = r^2$. An equation $x^2 + y^2 + ux + vy + w = 0$ is handled by completing both squares. If the squared radius obtained is negative, the set is empty; if it is zero, it is a point.

## Equation of a circle

$x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x - 2)^2 + (y + 3)^2 = 4$. Centre $(2, -3)$, radius 2. The signs of the coordinates of the centre are opposite to those appearing in the brackets.

# Equation of a circle

A quadratic equation in x and y is not always a circle: check the coefficients and the real radius.

# Equation of a circle

Recognise $x^2 + y^2 + 2x - 8y + 17 = 0$.

## Solving a configuration

An intersection is obtained by solving the equations simultaneously. For a circle and a line parallel to an axis, substituting the constant coordinate gives a quadratic equation. For a perpendicular bisector, write the equality of the squared distances; the quadratic terms cancel.

## Solving a configuration

The circle $x^2 + y^2 = 25$ and the line $x = 3$ meet at $(3, 4)$ and $(3, -4)$. The perpendicular bisector of $A(0, 0)$ and $B(4, 2)$ satisfies $x^2 + y^2 = (x - 4)^2 + (y - 2)^2$, so $2x + y = 5$.

## Solving a configuration

An intersection is not the union of the solution sets. Check the points in both equations.

## Solving a configuration

Intersect $(x-1)^2+(y+2)^2=9$ with $y=1$.